

AREQUIPA, Peru

WMO: 847520

Lat: 16.341S

Lon: 71.583W

Elev: 8405

StdP: 10.75

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	42.9	44.5	-9.3	4.6	51.9	-4.5	5.9	53.3	24.4	52.9	20.2	54.4	6.6	30	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	19.2	75.2	48.4	73.7	47.8	73.2	47.6	57.7	65.3	56.7	64.1	56.0	63.3	11.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.5	90.0	59.1	54.0	85.3	58.2	53.6	84.0	57.9	29.0	65.1	28.3	64.1	27.7	63.0	61.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.4	15.6	14.2	DB	38.5	77.5	2.6	1.8	36.7	78.8	35.2	79.8	33.7	80.8	31.8	82.1	(4)
(5)			WB	26.2	59.5	1.0	0.8	25.5	60.1	25.0	60.6	24.4	61.1	23.7	61.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.4	60.0	59.7	60.0	59.9	59.2	58.2	57.9	58.8	59.8	59.8	59.7	60.2
	DBStd	2.61	2.75	2.93	2.51	2.53	2.54	2.31	2.44	2.36	2.24	2.35	2.50	2.67
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	2038	156	149	157	152	181	203	219	193	157	162	158	151
	CDD50	3443	310	272	309	298	285	247	246	272	293	303	292	316
	CDD65	5	1	1	1	0	0	0	0	0	0	0	1	1
	CDH74	95	4	4	8	12	13	5	4	8	12	8	6	11
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.7	6.1	5.9	5.9	6.0	6.9	7.3	7.4	7.4	7.1	6.9	6.7	6.3
Precipitation	PrecAvg	3.4	0.9	1.2	0.9	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.2
	PrecMax	9.3	4.7	4.8	4.3	0.2	0.1	0.0	0.0	1.5	0.7	0.1	0.4	0.8
	PrecMin	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.7	1.0	1.3	1.1	0.0	0.0	0.0	0.0	0.3	0.2	0.0	0.1	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.0	74.2	75.1	75.5	75.5	74.9	73.8	75.2	75.5	75.2	75.1	75.4
		MCWB	52.3	55.7	52.8	49.8	47.6	46.1	44.8	44.8	47.9	48.5	48.0	49.2
	2%	DB	71.9	71.9	72.0	73.4	73.5	72.9	71.9	73.2	73.5	73.4	73.3	73.4
		MCWB	52.3	55.9	54.7	49.6	47.2	45.9	44.4	44.5	45.8	47.5	47.2	49.4
	5%	DB	70.1	69.9	70.2	71.6	71.7	71.2	70.2	71.5	71.9	71.8	71.6	71.5
		MCWB	52.3	55.1	54.2	50.1	46.4	45.2	44.0	44.4	45.6	46.2	47.1	49.3
	10%	DB	68.2	67.8	68.2	69.7	69.8	69.4	69.2	69.8	70.2	70.1	69.8	69.6
		MCWB	52.5	54.6	53.9	50.6	45.9	44.1	43.6	43.7	45.2	45.7	46.7	49.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.4	58.9	58.5	56.5	53.0	51.2	48.7	47.3	50.3	51.6	52.7	55.4
		MCDB	65.6	66.5	67.2	65.9	63.4	67.2	65.6	68.3	69.0	67.3	66.1	63.7
	2%	WB	56.9	57.9	57.4	55.2	51.1	47.8	46.7	45.8	48.8	49.5	51.2	53.8
		MCDB	63.8	65.3	65.5	64.7	62.9	66.0	65.9	67.7	67.7	66.8	64.8	63.1
	5%	WB	55.9	56.9	56.4	54.1	49.6	46.4	45.3	44.9	47.4	48.2	49.9	52.6
		MCDB	63.0	63.9	64.3	63.9	63.7	65.4	65.7	67.3	67.1	66.0	64.6	62.7
	10%	WB	54.9	56.1	55.6	53.1	48.6	45.2	44.3	44.1	46.1	46.8	48.9	51.6
		MCDB	62.1	62.8	63.3	63.1	64.2	64.9	65.4	66.8	66.4	65.4	64.3	62.4
Mean Daily Temperature Range	5% DB	MDBR	16.9	16.0	17.5	20.3	22.5	23.4	23.3	23.3	22.6	22.0	21.2	19.2
		MCDBR	19.4	19.2	20.3	22.5	24.2	25.2	24.7	24.9	24.0	23.6	22.6	21.4
		MCWBR	8.4	7.7	8.4	10.4	12.0	11.9	11.7	12.1	11.7	11.0	10.7	9.5
	5% WB	MCDBR	14.2	14.1	15.1	17.2	20.2	21.2	21.1	21.7	20.7	19.7	18.9	16.8
		MCWBR	7.0	7.0	7.5	9.0	11.6	11.4	11.0	11.5	10.9	10.8	10.0	8.4
Clear-Sky Solar Irradiance	taub	0.423	0.412	0.403	0.390	0.321	0.304	0.300	0.337	0.371	0.376	0.396	0.415	
	taud	2.117	2.184	2.218	2.213	2.376	2.443	2.449	2.306	2.195	2.179	2.111	2.097	
	Ebn at Noon	293	293	289	281	291	291	296	293	294	300	299	295	
	Edh at Noon	54	50	47	44	35	31	32	39	47	49	54	55	
All-Sky Solar Radiation	RadAvg	2102	1905	1972	1950	1849	1725	1766	2005	2240	2441	2575	2372	
	RadStd	213	202	163	108	69	64	81	74	95	84	92	131	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	-1.12	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page